

SOC HOME LAB

THREAT DETECTION & INCIDENT RESPONSE REPORT

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Role: SOC Analyst (Home Lab Project)

Environment: Wazuh SIEM, Sysmon, Windows 10, Ubuntu Server, Kali Linux

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Executive Summary

This project demonstrates the deployment and operation of a Security Operations Center (SOC) Home Lab designed to simulate real-world threat detection, investigation, and incident response activities.

The environment consisted of a Windows 10 endpoint monitored by Sysmon and Wazuh Agent, a centralized Wazuh SIEM deployed on Ubuntu Server, and a Kali Linux system used to simulate attacker behavior.

Multiple security events were generated, collected, analyzed, and investigated through Wazuh SIEM. The project validated the ability to detect suspicious PowerShell activity, failed authentication attempts, account creation events, and reconnaissance activities while leveraging the MITRE ATT&CK framework for threat classification.

The objective was to develop practical SOC analyst skills including security monitoring, alert triage, threat hunting, event correlation, and incident documentation.

Project Objectives

The primary objectives of this project were:

- Deploy a functional SOC environment
- Configure centralized log collection
- Integrate Sysmon endpoint telemetry
- Generate security-relevant events
- Investigate alerts using Wazuh SIEM
- Perform threat hunting activities
- Correlate events with MITRE ATT&CK
- Develop incident response workflows

Lab Architecture

Infrastructure Overview

System	Role	IP Address
Kali Linux	Attack Simulation	192.168.207.129
Ubuntu Server	Wazuh SIEM	192.168.207.128
Windows 10	Monitored Endpoint	192.168.207.130

Security Monitoring Workflow

Attacker Activity (Kali Linux)

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Windows Endpoint Generates Events

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Sysmon Records Telemetry

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Wazuh Agent Collects Logs

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Wazuh Manager Processes Events

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Threat Detection Rules Trigger

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SOC Investigation Begins

Tools & Technologies

Security Monitoring

- Wazuh SIEM
- Wazuh Dashboard
- Wazuh Agent

Endpoint Telemetry

- Microsoft Sysmon

Operating Systems

- Ubuntu Server
- Windows 10
- Kali Linux

Frameworks

- MITRE ATT&CK

Virtualization

- VMware Workstation

Detection Use Cases

Detection 1: PowerShell Execution Activity

Objective

Detect potentially suspicious PowerShell activity frequently associated with attacker execution techniques.

Detection Source

- Sysmon Event ID 1
- Process Creation Events

Wazuh Detection

Rule ID:

92027

Rule Description:

PowerShell Process Spawned PowerShell Instance

MITRE ATT&CK Mapping

Technique:

T1059.001 – PowerShell

Tactic:

Execution

Analyst Assessment

PowerShell activity was successfully detected and logged by Wazuh SIEM. This behavior is commonly associated with script execution, malware delivery, and post-exploitation activities.

Security Impact

Medium

Detection 2: Failed Authentication Attempts

Objective

Detect unsuccessful login attempts that may indicate brute-force activity or unauthorized access attempts.

Detection Source

Windows Security Logs

Event ID

4625

MITRE ATT&CK Mapping

Technique:

T1110 – Brute Force

Tactic:

Credential Access

Analyst Assessment

Multiple failed authentication attempts were successfully logged and monitored.

Security Impact

Medium

Detection 3: User Account Creation Activity

Objective

Monitor administrative account creation events.

Event ID

4720

MITRE ATT&CK Mapping

Technique:

T1136 – Create Account

Tactic:

Persistence

Analyst Assessment

Account creation activity was detected and correlated through centralized logging.

Security Impact

Medium

Detection 4: Network Reconnaissance

Objective

Simulate attacker reconnaissance activity using Nmap.

Tool

Nmap

Commands Executed

```
nmap -sS 192.168.207.130
```

```
nmap -A 192.168.207.130
```

MITRE ATT&CK Mapping

Technique:

T1595 – Active Scanning

Tactic:

Reconnaissance

Analyst Assessment

Reconnaissance activity was generated from the attacker workstation and documented as part of the threat simulation process.

Security Impact

Low

Threat Hunting Investigation

Investigation Overview

Threat hunting activities were performed using the Wazuh Threat Hunting module to review endpoint telemetry and identify suspicious behaviors.

The investigation included:

- Alert validation
 - Event correlation
 - Timeline analysis
 - Process review
 - MITRE ATT&CK classification
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Investigation Findings

The environment successfully generated actionable telemetry that enabled analysts to:

- Investigate PowerShell execution activity
- Review authentication events
- Monitor account management actions
- Analyze endpoint logs
- Validate detection coverage

No malicious compromise was identified.

All activities originated from controlled testing scenarios.

MITRE ATT&CK Mapping

Detection	Technique	Description
PowerShell Execution	T1059.001	PowerShell
Failed Logins	T1110	Brute Force
User Creation	T1136	Create Account
Network Reconnaissance	T1595	Active Scanning

Incident Response Workflow

The following workflow was followed during each investigation:

1. Alert Generation
2. Initial Triage
3. Event Validation
4. Event Correlation

5. Threat Classification
 6. MITRE ATT&CK Mapping
 7. Documentation
 8. Remediation Recommendations
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Lessons Learned

This project improved practical knowledge in:

- Security Monitoring
 - SIEM Administration
 - Threat Hunting
 - Log Analysis
 - Endpoint Detection
 - Alert Triage
 - Incident Response
 - MITRE ATT&CK Mapping
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Recommendations

Future improvements include:

- Active Directory Integration
 - Suricata IDS Deployment
 - Sigma Rule Integration
 - Threat Intelligence Feeds
 - Multi-Endpoint Monitoring
 - Automated Incident Response
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Conclusion

This project successfully demonstrated the deployment and operation of a SOC monitoring environment capable of collecting endpoint telemetry, generating security alerts, conducting threat hunting investigations, and supporting incident response activities.

The lab provided hands-on experience aligned with real-world SOC Analyst responsibilities and strengthened practical skills in detection engineering, alert analysis, and security operations.

The completed environment serves as a foundational platform for advanced Blue Team, Threat Hunting, and Cloud Security projects.